INCH-POUND

MIL-C-85049/41A
30 November 1990
SUPERSEDING
MIL-C-85049/41
10 January 1980
MS3057E
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, NONENVIRONMENTAL, STRAIN RELIEF, STRAIGHT, CATEGORY 4C (FOR MIL-C-5015 SOLDER TYPE, V THREAD OF MS310X CLASSES A, B, C OR K CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

INACTIVE FOR NEW DESIGN EQUIPMENT OR MODIFICATION OF EXISTING EQUIPMENT, USE MIL-C-85049/42.

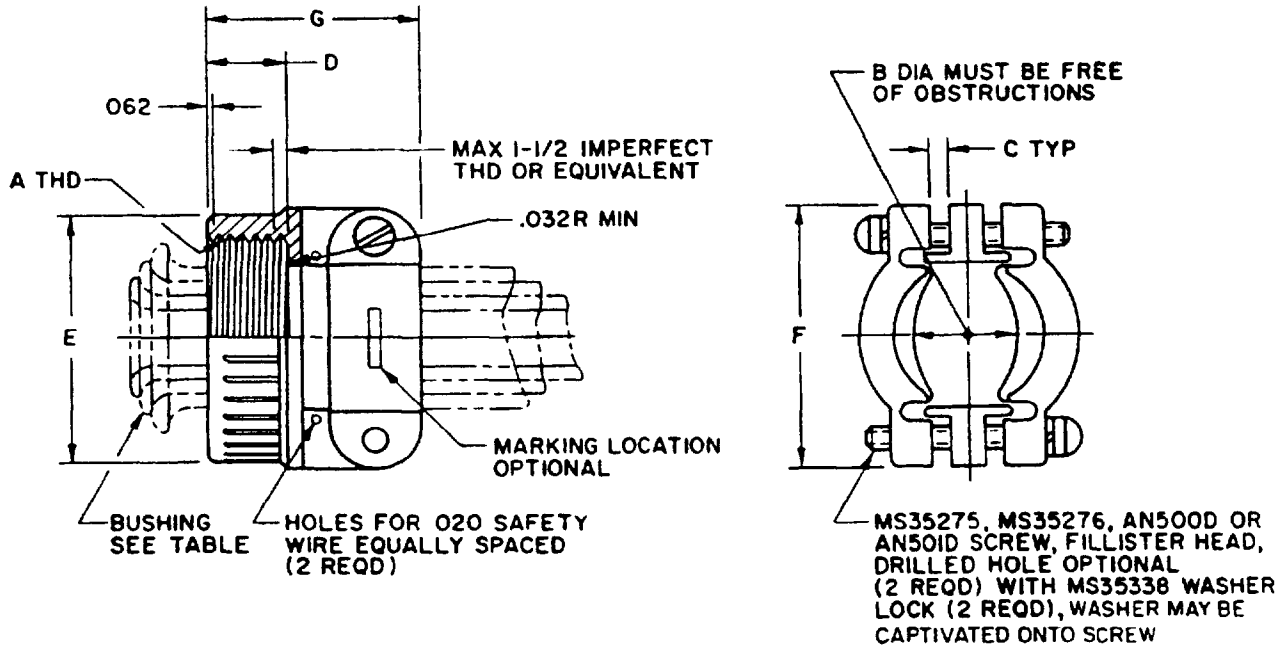


FIGURE 1. Configuration and dimensions.

DISTRIBUTION STATEMENT A: Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensions.

Dash No.	Connector Shell Size (Ref) See Note 7	A Thread	B Wire Bundle Accommodation Max	C	D	E ± .031 Dia	F ± .031	G ±.031	Bushing (Ref) See Note 8
-3A	8S & 10S	.500-28 UNEF-2B	.250	.047	.406	.688	.812	.812	MS3420-3
-4A	10SL, 12S & 12	.625-24 UNEF-2B	.312	.062		.812	.875		MS3420-4
-6A	14S & 14	.750-20 UNEF-2B	.438	.078		.969	1.062	.875	MS3420-6
-8A	16S & 16	.875-20 UNEF-2B	.562	.109		1.094	1.156	.938	MS3420-8
-10A	18	1.000-20 UNEF-2B	.625	.125		1.188	1.250		MS3420-10
-12A	20 & 22	1.188-18 UNEF-2B	.750	.156		1.375	1.469		MS3420-12
-16A	24 & 28	1.438-18 UNEF-2B	.938	.188		1.656	1.688	1.031	MS3420-16
-20A	32	1.750-18 UNS-2B	1.250	.250	.469	2 031	2.031	1 094	MS3420-20
-24A	36	2.000-18 UNS-2B	1.375	.297	.531	2.219	2.281	1.156	MS3420-24
-28A	40	2.250-16 UN-2B	1 625	.328		2.500	2.688	1.688	MS3420-28
-32A	44	2.500-16 UN-2B	1.875	.375	.594	2.781	2.938	1.750	MS3420-32
-40A	48	3.000-16 UN-2B	2.375	.453	.656	3.281	3.500		MS3420-40

Dash No.	Fillister Head Screw				Screw Size	MS35338 Washer, Lock Dash Number
	AN500D Dash Number	MS35275 Dash Number	AN501D Dash Number	MS25276 Dash Number		
-3A	4-7	216			4-40	135
-4A	4-10	218			4-40	135
-6A	6-12	232			6-32	136
-8A	6-12	232			6-32	136
-10A	6-14	248			6-32	136
-12A	8-16	249			8-32	137
-16A	8-18	N/A			8-32	137
-20A	8-24	251			8-32	137
-24A			10-24	269	10-32	138
-28A	416-28	286			1/4-20	139
-32A	416-32	287			1/4-20	139
-40A	416-48	291			1/4-20	139

N/A = Not Availabl.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

INCH	mm	INCH	mm	INCH	mm
0.016	0.41	0.453	11.51	1.250	31.75
0.020	0.51	0.469	11.91	1.375	34.93
0.031	0.79	0.500	12.70	1.438	36.53
0.032	0.81	0.531	13.49	1.469	37.31
0.047	1.19	0.562	14.27	1.625	41.28
0.062	1.57	0.594	15.09	1.688	42.88
0.078	1.98	0.625	15.88	1.750	44.45
0.109	2.77	0.656	16.66	1.875	47.63
0.125	3.18	0.750	19.05	2.000	50.80
0.156	3.96	0.812	20.62	2.031	51.59
0.188	4.78	0.875	22.22	2.250	57.15
0.250	6.35	0.938	23.83	2.281	57.94
0.297	7.54	1.000	25.40	2.375	60.32
0.312	7.92	1.031	26.19	2.500	63.50
0.328	8.33	1.062	26.97	2.688	68.28
0.375	9.53	1.094	27.79	2.938	74.63
0.406	10.31	1.156	29.36	3.000	76.20
0.438	11.13	1.188	30.18	3.500	88.90

REQUIREMENTS:

1. Design and construction:

   Dimensions and configuration: See figure 1.

2. Material and finish: See table II.

TABLE II. Material and finish.

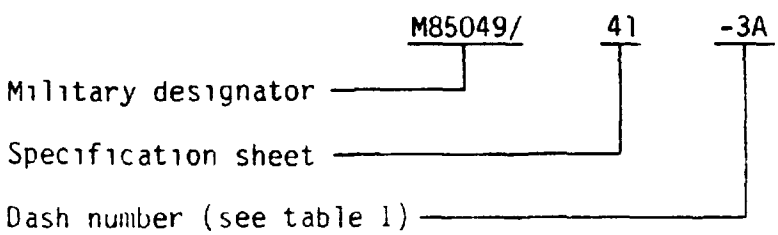
Material	Finish
Aluminum alloy in accordance with MIL-C-85049	1/ W

1/ W finish is not for use in space application.

3. Qualification: See MIL-C-85049, category 4C, examination of product, salt spray and post examination, only tests required.

   Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

4. Part Number: See example below:



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

6. Supersession data:

MS3057 - * * M85049/41 - *

Dash Number

Finish

Old	New
A	W
C	W
N	W

NOTES:

1. Dimensions are in inches.

2. Metric equivalents are based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.

3. Unless otherwise specified, tolerances shall be as follows: Decimals ±.016, angles ±2°.

4. Dimensions apply after plating.

5. Dimensions and tolerancing in accordance with ANSI Y14.5.

6. Remove all burrs and sharp edges.

7. Clamp mates with MIL-C-5015 Solder Type, MS310X MS "V" thread class A, B, C and K connectors.

8. Bushings for reference only and not supplied as part of clamp. For additional closedown of the B dimension, bushing MS3420-3 through MS3420-40 may be used. One bushing as required for M85049/41-3 through M85049/41-12, corresponding to MS3420-3 through MS3420-12. Two bushings as required for M85049/41-16 through M85049/41-24, one each corresponding to MS3420-16 through MS3420-24, and MS3420-12 through MS3420-20. (Examples: M85049/41-16 requires one MS3420-16 and one MS3420-12.) Three bushings as required for M85049/41-28 through M85049/41-40, one each corresponding to MS3420-28 through MS3420-40, MS3420-24 through MS3420-32, and MS3420-20 through MS3420-28. (Examples: M85049/41-40 requires one MS3420-40, one MS3420-32, and one MS3420-28.)

9. The B dia dimension is meant to define the maximum wire bundle outside diameter which can be accommodated by the clamp. This dimension is not meant to define the clamp hardware dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 99
DLA - ES

Preparing activity:
Navy - AS
(Project No. 5935-3729-15)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.